Printed Pages:2 **Sub Code: KEE-101T**

Paper Id: 233106 **Roll No.** | | | | | | | | | | |

B.TECH
(SEM I) THEORY EXAMINATION 2022-23
BASIC ELECTRICAL ENGINEERING

Time: 3 Hours ***Total Marks: 100***

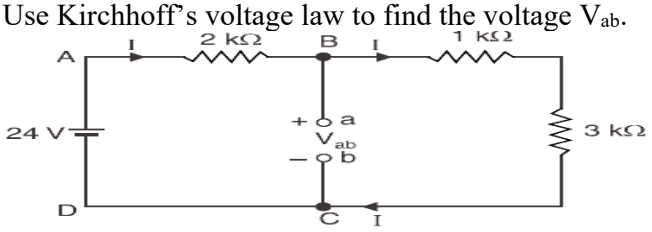
Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **10 x 2 = 20**

(a) Explain the advantage of Nodal analysis over Mesh analysis.

(b) Use Kirchhoff's voltage law to find the voltage V_{ab}.



(c) A coil having a resistance of 7Ω and an inductance of 31·8 mH is connected to 230 V, 50 Hz supply. Calculate power factor

(d) Draw the Impedance triangle of inductive circuit.

(e) A magnetic circuit has m.m.f. of 400 AT andreluctance of 2×10^5 AT/Wb. Find out magnetic flux in the magnetic circuit.

(f) Write advantages of Auto-transformer over single phase transformer.

(g) State the charactertics of synchronous motor.

(h) A three phase, 50 Hz induction motor has a 4 poles full load speed of 1440 rpm. Find out full load slip.

(i) A ceiling fan of 200 watt is used for four hours per day, and for 120 days per year, what would be the annual energy consumption? If the price per kWh for electricity is Rs 6, what is the annual cost to operate the ceiling fan?

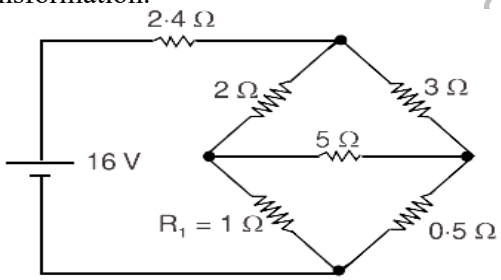
(j) What is the necessity of earthing?

\

SECTION B

2. Attempt any *three* of the following: **3x10=30**

(a) Find out total current drawn from the voltage source (16 v battery), By Star/Delta transformation.



(b) Find the average value, r.m.s. value, form factor and peak factor for halfwave rectified alternating current.

(c) Draw the Equivalent circuit of the transformer, referred to secondary side. The equivalent parameters of a transformer, having a turns ratio of 5, are R_1=0.5Ω, R_2=0.021Ω, X_1=3.2Ω, X_2=0.12Ω.Determine Equivalent resistance, reactance and impedance referred to secondary side, and total copper loss .

QP23DP2_311 | 22-03-2023 09:06:14 | 115.243.172.58

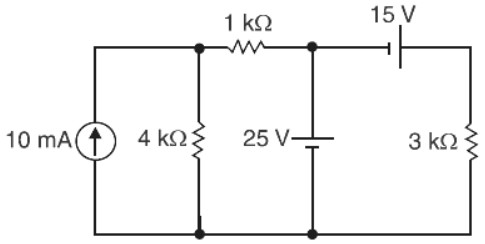
(d) Explain the basic principle of working of DC Machine and drive the back EMF equation.

(e) Explain the following characteristics for Battery: Battery Backup time, Ampere-hour efficiency,watt-hour efficiency,Service life.

SECTION C

3. Attempt any *one* part of the following: 1x10=10

(a) Using the superpositionprinciple, find the voltage across 1k Ω resistor



(b) Explain in detail the procedure for finding the Thevenin's equivalent circuit.

4. Attempt any *one* part of the following: 1x10=10

(a) Prove that that power absorbed in a pure capacitance is zero. A 318μF capacitor is connected to 230V, 50 Hz find out capacitive reactance, rms value of current.

(b) A choke coil takes a current of 2A lagging 60° behind the applied voltage of 200 V at 50Hz.
(i) Calculate impedance, resistance and inductance of the coil.
(ii) Also find the power consumed when the coil is connected across 100 V, 25 Hz supply.

5. Attempt any *one* part of the following: 1x10=10

(a) Explain in detail the properties of following magnetic material with examples: diamagnetic, paramagnetic, ferromagnetic material.

(b) Explain in detail various three-phase transformer connections.

6. Attempt any *one* part of the following: 1x10=10

(a) A 25kW, 250-V, DC shunt motor has armature and field resistances of 0.06 Ω and 100 Ω respectively. Determine the total armature power developed when working asi. agenerator delivering 25kW as output ii. a motor taking 25 kW input

(b) Explain Working Principle of Operation of three phase induction motor and compare squirrel cage induction motor withslip-ring induction motor.

7. Attempt any *one* part of the following: 1x10=10

(a) Explain working principle of miniature circuit breaker (MCB) with the help of diagram and compare with ELCB.

(b) Explain in detail different types of Insulation used in wire or cable.

QP23DP2_311 | 22-03-2023 09:06:14 | 115.243.172.58

QP23DP2_311 | 22-03-2023 09:06:14 | 115.243.172.58